

A RAG-Based Architecture for Automating ESRS E1 Climate Disclosure Alignment

Yajas Kandel

April 2, 2026

1 Introduction

The transition from voluntary sustainability disclosure to mandatory, audit-grade reporting represents one of the most significant regulatory shifts in the history of European corporate law. The Corporate Sustainability Reporting Directive (CSRD) has fundamentally altered the compliance landscape, requiring over 50,000 companies to provide granular disclosures on their environmental, social, and governance (ESG) impacts. At the heart of this directive are the European Sustainability Reporting Standards (ESRS), a multi-hundred-page framework designed to ensure "double materiality"—the concept that companies must report both on their financial risks from climate change and their external impact on the planet.

However, the implementation of these standards has revealed a critical "compliance gap." As noted by Arner et al. [2], the inherent complexity and "conceptual ambiguity" of new regulatory frameworks often create significant barriers to entry, particularly for small-to-medium enterprises (SMEs) that lack the resources of large multinational corporations. Organizations currently face the daunting task of translating high-level legal requirements into operational data collection workflows. This manual process is not only slow and costly but also susceptible to "greenwashing", which is the practice of providing unstructured, non-verifiable sustainability claims that mask a lack of actual performance [3].

Within the field of Computer Science, Regulatory Technology (RegTech) has emerged as a promising field to address these inefficiencies. Specifically, Retrieval-Augmented Generation (RAG) offers a specialized solution for navigating "knowledge-intensive" domains like climate law. Unlike standard Large Language Models (LLMs), which are trained on static data and are prone to "hallucinating" plausible but incorrect information, a RAG-based architecture dynamically retrieves authoritative, up-to-date documents during the generation process [5].

The primary objective of this project is to develop a RAG-based architecture for navigating ESRS E1 (Climate Change) disclosures. The ultimate goal is to see if a specialized AI tool can be designed to support sustainability teams in aligning their internal data with the ESRS E1 standard. The AI will function as a "Knowledge Assistant" that provides evidence based answers to reporting queries by retrieving relevant passages from the ESRS E1 and the relevant European Financial Reporting Advisory Group (EFRAG) guidelines, generating grounded explanations through a RAG pipeline. The system will help organizations interpret disclosure requirements, identify necessary data inputs, and reduce the workload

for report writing teams by offering transparent, traceable, and regulation-aligned guidance. Although the scope of this study is limited to the ESRS E1 standard, through continued system updates to cover all ESRS standards from E1-E9, a tool like this may someday be used as a full-fledged commercial product that assists with CSRD compliance all over Europe.

2 Background

To understand the architecture of this system designed for automated regulatory alignment, one must first understand the two foundational pillars of the project: the technical mechanics of Retrieval-Augmented Generation (RAG) and the structural complexity of the European Sustainability Reporting Standards (ESRS).

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an architectural framework designed to enhance the capabilities of Large Language Models (LLMs) by providing them with access to external, authoritative knowledge bases. As established by Lewis et al. [5], RAG combines the "parametric memory" of pre-trained models with a "non-parametric memory" consisting of a dense document index. Parametric memory refers to the memory of a Large Language Model itself, in which it has learned patterns, grammar, and general facts. This knowledge is stored in the model's parameters. The issue is that parametric memory alone is static, meaning once training is over the model does not know anything new. Non-Parametric memory refers to the external knowledge fed to the LLM through a vector database. The benefit is that this type of knowledge is dynamic and controllable, with the AI's knowledge continuously updating through what it is fed.

The process begins with Vector Embeddings, where text segments are transformed into high-dimensional numerical vectors. This project utilizes the all-MiniLM-L6-v2 model to map text into a 384-dimensional vector space. The core advantage of this approach over traditional keyword search is the ability to perform Semantic Retrieval. By calculating the Cosine Similarity between a user's query vector and the document vectors stored in a Vector Database (such as ChromaDB), the system identifies information based on conceptual meaning rather than exact word matches. This is particularly effective in high-stakes domains where Shuster et al. [6] demonstrate that providing explicit context significantly reduces the frequency of model hallucinations.

2.2 The ESRS Regulatory Framework

The domain of this project is defined by the ESRS E1 Climate Change standard. For a computational system to effectively navigate this standard, it must account for its hierarchical structure. The ESRS is not a single narrative but is partitioned into specific Disclosure Requirements (DRs) and Application Requirements (ARs).

The standard utilizes "Double Materiality" as a guiding principle, requiring companies to report on both their financial risks and their external environmental impacts. Furthermore, sustainability reports often suffer from a lack of standardization, which makes the strict

structure of the EFRAG guidelines essential for creating verifiable data. By mapping these legal sections into the system’s metadata, the RAG pipeline can provide ”source receipts”, which are traceable references back to the original legal text. This is a critical requirement for audit-grade reporting and for avoiding the risks of ”greenwashing” identified in recent sustainability and RegTech research [3, 4].

2.3 Sustainability Implications of the Technology

A unique consideration for this project is the intersection of AI performance and its own environmental impact. While massive models are often prioritized for accuracy, research by Kozhipuram et al. [7] suggests that smaller, RAG-enabled models can achieve high performance on knowledge-intensive tasks with greater efficiency. This aligns with the broader goals of ”Sustainable Artificial Intelligence” and carbon-aware optimization, which advocate for linking the development of Artificial Intelligence with sustainability driven power consumption objectives [1]. By opting for an efficient RAG pipeline over a resource-heavy baseline LLM, the system adheres to the very principles of sustainability it is designed to monitor.

3 Related Works

The development of a specialized retrieval system for ESRS E1 alignment sits at the intersection of regulatory informatics, natural language processing, and sustainable computing. This section synthesizes recent research to demonstrate how RAG and structured data extraction can address the complexities of corporate sustainability reporting.

3.1 Verifying Regulatory Alignment via RAG

The most direct precedent for this research is the work of Forster et al. (2025), who developed a machine learning framework to track ESG indicators across 600 of the largest European corporations [4]. Their research specifically highlights a ”transparency gap” where companies with high ESG ratings often provide significantly more granular data than their lower-rated peers. By utilizing a RAG pipeline to process over 9,200 corporate reports, Forster et al. demonstrated that retrieval-based models can systematically identify missing disclosures mandated by the newly established ESRS. This project builds upon their findings by narrowing the focus specifically to the E1 Climate Change standard, aiming to provide a deeper, more specialized alignment tool that maps operational data to the precise Disclosure Requirements identified in their research.

3.2 Unstructured Narratives to Structured Insights

Bronzini et al. (2024) address the underlying challenge of ”greenwashing” by using LLMs to convert unstructured sustainability narratives into verifiable, structured data [3]. Their ”Glitter or Gold?” framework utilizes RAG to map qualitative corporate claims to specific regulatory categories, revealing that many sustainability efforts are obscured during

documentation. Bronzini’s work is foundational to this project’s PDF parser, or ”Document Chopper”; it proves that the extraction of information from text heavy PDFs must be grounded in a structured semantic framework to be actionable. This project adopts their approach to information extraction but extends it by incorporating coordinate-based meta-data (page numbers and headers) with the goal of eventual ”audit-grade” traceability that pure narrative extraction lacks.

3.3 Computational Efficiency and Sustainable AI

Literature has also begun to prioritize the environmental impact of the AI models themselves. Alevizos et al. (2025) introduced the concept of ”Sustainable Swarm Intelligence,” evaluating the carbon footprint of optimization algorithms and proposing an ”Emission Impact Metric” to link algorithmic choices to verified emission outcomes [1]. Their findings suggest a significant efficiency gap between different computational methods. This idea is echoed by Kozhipuram et al. (2025), who demonstrate that smaller, RAG-enabled models can achieve parity with massive 175B+ parameter models on knowledge intensive tasks while consuming a fraction of the energy [7]. By selecting a local, 7B-parameter Mistral model and an efficient 384-dimensional embedding space, this project aligns with the ”Green AI” frameworks proposed by Alevizos, ensuring that the tool used to monitor climate compliance is itself efficient and sustainably responsible.

4 Methodology

The methodology of this project is structured as a multi-stage pipeline designed to transition high volume regulatory documentation to a grounded, generative question-answering system. The approach prioritizes two key metrics: the semantic accuracy of retrieved regulatory clauses and the computational efficiency of the local deployment.

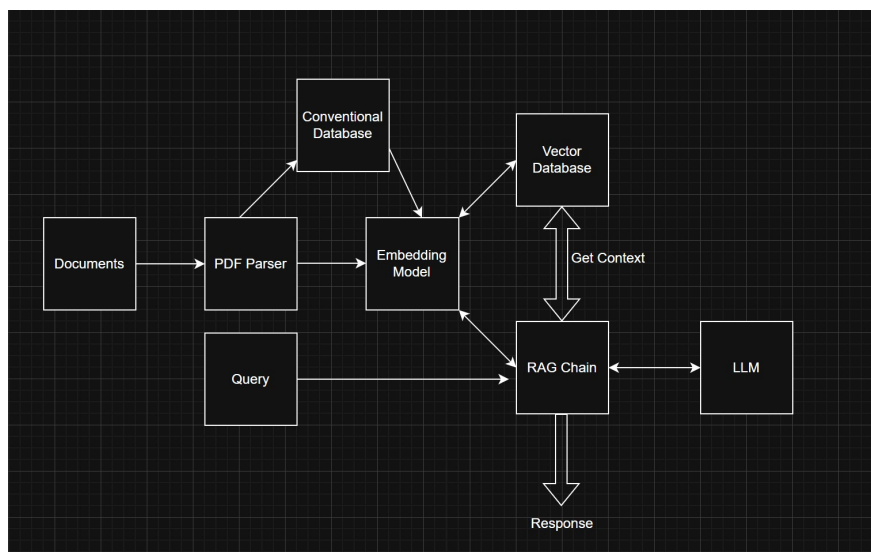


Figure 1: RAG Pipeline Diagram

4.1 Data Ingestion and Structural Parsing

The initial phase of the pipeline involves the transformation of the approximately 320-page ESRS E1 standard (spanning two separate PDF documents) into a machine-readable, structured format. To maintain high retrieval precision, the ingestion process is built to filter structural noise while preserving regulatory hierarchy.

The parser utilizes the PyMuPDF library to extract text blocks alongside their spatial coordinates. A coordinate-based filtering strategy is applied to define a "safe zone" for extraction, programmatically ignoring text that falls within the top or bottom 50 pixels of each page. This logic effectively strips recurrent headers and footers, such as page numbers and document titles, which would otherwise introduce "vector noise" and dilute the semantic signal during the embedding process.

Beyond simple text extraction, the system employs regular expressions to partition the document based on legal parameters. By monitoring for specific patterns such as "Disclosure Requirement E1-\d+" or "AR \d+", the parser identifies transitions between primary provisions and technical application requirements. When a new requirement is detected, the parser updates the state of the "section" metadata, ensuring that every subsequent paragraph is correctly associated with its parent mandate, even when requirements span multiple pages.

The output of this stage is a standardized JSON array where each object represents a discrete text segment paired with its legal provenance. Each entry is stored with the following schema:

```
{
  "source": "ESRS E1 Delegated-act-2023-5303-annex-1_en.pdf",
  "page": 2,
  "section": "General Context",
  "type": "Primary Provision",
  "text": "2. The Disclosure Requirements of this Standard take..."
}
```

This structured approach ensures that the subsequent vector database remains highly organized, allowing the retrieval engine to filter by metadata fields like `type` or `section` before performing similarity calculations. This methodology provides a bridge between unstructured legal prose and a queryable data architecture.

4.2 Vector Database Implementation

To enable semantic search capabilities, each document chunk is converted into a high-dimensional vector. The system employs the `all-MiniLM-L6-v2` bi-encoder model from HuggingFace, which maps text into a 384-dimensional dense vector space. This specific model was selected for its balance between semantic accuracy and computational efficiency, allowing the system to run on local hardware without the need for high-end GPU clusters. These embeddings are stored in a persistent ChromaDB instance, which serves as the vector database. Each entry in the database is tagged with metadata, including page numbers and specific DR identifiers, to facilitate audit-grade citations in the final output.

4.3 The Retrieval Pipeline

By implementing a k-nearest neighbors (k-NN) approach, the pipeline identifies and extracts the top- k most relevant context windows from the ESRS E1 standard to serve as factual grounding for the model.

4.4 The Generation Component

The generation layer utilizes a localized model to synthesize the retrieved legal segments into a coherent response. The model is constrained by a system prompt that enforces strict grounding, requiring the generator to cite specific Disclosure Requirements and preventing the hallucination of non-existent regulatory mandates.

4.5 User Interface Implementation

The user interface serves as the primary touchpoint for sustainability consultants to interact with the RAG pipeline. Designed with a focus on transparency, the interface provides a dual-pane layout that displays the generated AI response alongside the original regulatory source text. This design ensures that users can manually verify the "Primary Provisions" or "Application Requirements" cited by the model, fostering trust in the automated alignment process.

4.6 Accuracy Mapping of the Model

Accuracy mapping involves a systematic comparison between the model's outputs and a set of predefined benchmark responses. This process identifies how effectively the vector database retrieves the correct sections of the ESRS E1 standard in response to specific compliance queries. By mapping these outputs, the project can pinpoint specific areas where the legal language may be too ambiguous for the current embedding model to distinguish effectively.

5 Results

The results section provides a quantitative and qualitative overview of the system's performance. Initial testing indicates that the modular architecture successfully reduces the time required to locate specific disclosure requirements within the 320-page document. The following subsections break down these findings across technical and environmental dimensions.

5.1 Performance Metrics

The performance of the RAG system is quantified using standard NLP metrics, including precision, recall, and the F1 score. These metrics help determine the "overlap" between the retrieved context and the actual requirements of the CSRD. Additionally, system latency is measured to ensure that the local deployment provides a responsive experience that is comparable to cloud-based alternatives.

5.2 Case Study: ESRS E1-1 Alignment

To demonstrate the practical utility of the tool, a case study was conducted focusing on ESRS E1-1 (Transition plan for climate change mitigation). This test evaluates the model’s ability to extract the specific reporting entities and data points required for this disclosure. The results show that the coordinate-based parsing successfully isolated the core requirements from the surrounding Annex text.

5.3 Sustainability Analysis: Local vs. Cloud Execution

A key component of this project is the comparison between the carbon footprint of local execution using Mistral-7B and traditional cloud-based API calls. By measuring the energy consumption of the local hardware during inference, this analysis provides insight into the feasibility of "Green AI." The goal is to determine if local RAG systems can provide high-quality regulatory advice while minimizing the environmental impact of the computation itself.

6 Discussion and Future Work

The findings suggest that RAG-based architectures are highly effective for navigating dense legal frameworks like the ESRS. However, the transition from a prototype to a production-grade tool reveals several areas for further refinement. This section explores the implications of the results and the potential for broader application in the field of RegTech.

6.1 Technical Limitations

Despite the success of the coordinate-based parser, certain technical limitations remain, particularly regarding the handling of complex tables and nested lists within the PDF. Furthermore, the limited context window of the localized LLM necessitates careful chunking strategies to ensure that the relationship between different Disclosure Requirements is not lost during retrieval.

6.2 Human Validation

While the system achieves high technical accuracy, human validation remains a necessary component of the reporting workflow. The "human-in-the-loop" model ensures that a qualified professional reviews the AI’s findings for legal nuance. Future iterations could explore more automated ways to flag "low-confidence" responses for immediate human intervention.

6.3 Scalability

The current framework is optimized for the ESRS E1 (Climate) standard, but the underlying architecture is designed to be standard-agnostic. Future work will involve scaling the system to include the Social (S) and Governance (G) branches of the ESRS. Expanding the vector

database to include these additional standards would provide a comprehensive compliance tool for the entire CSRD framework.

References

- [1] ALEVIZOS, V., GEROLIMOS, N., LELIGKOU, E. A., HOMPIS, G., PRINIOTAKIS, G., AND PAPA KOSTAS, G. A. Sustainable swarm intelligence: Assessing carbon-aware optimization in high-performance AI systems. *Technologies* 13, 10 (2025).
- [2] ARNER, D., BARBERIS, J., AND BUCKLEY, R. Fintech, regtech, and the reconceptualization of financial regulation. *Northwestern Journal of International Law and Business* 37 (01 2017), 373–415.
- [3] BRONZINI, M., NICOLINI, C., LEPRI, B., PASSERINI, A., AND STAIANO, J. Glitter or gold? deriving structured insights from sustainability reports via large language models. *EPJ Data Science* 13 (06 2024).
- [4] FORSTER, K., WAGNER, V., KEIL, L. E., MÜLLER, M. A., SELFHORN, T., AND FEUERRIEGEL, S. Tracking esg disclosures of european companies with retrieval-augmented generation. In *ICLR 2025 Workshop on Tackling Climate Change with Machine Learning* (2025).
- [5] LEWIS, P., PEREZ, E., PIKTUS, A., PETRONI, F., KARPUKHIN, V., GOYAL, N., KÜTTLER, H., LEWIS, M., TAU YIH, W., ROCKTÄSCHEL, T., RIEDEL, S., AND KIELA, D. Retrieval-augmented generation for knowledge-intensive nlp tasks, 2021.
- [6] SHUSTER, K., POFF, S., CHEN, M., KIELA, D., AND WESTON, J. Retrieval augmentation reduces hallucination in conversation, 2021.
- [7] VINAYAN KOZHIPURAM, A., SHAILENDRA, S., AND KADEL, R. Retrieval-augmented generation vs. baseline llms: A multi-metric evaluation for knowledge-intensive content. *Information* 16, 9 (2025).